

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

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ABSTRACT

A frozen CLIP region can pass locality and mean-foil screens while responding more strongly to a non-target foil. We audit the maximum-target-drop single region from CCI candidates using matched prompt-mean and aggregate worst-foil margins. In the frozen archive, on COCO/OpenAI B/16, 41.18% of eligible images pass both screens yet have negative worst-foil margin; 64.38% [63.67,65.07]% of screen-passers fail. Selection-stage target-tolerant reranking at $\epsilon = .02$ switches 2.0–2.6% of images and improves normalized aggregate margin by .0150/.0108/.0118/.0087 across COCO/VOC and B/16/B/32. All specificity-gain 10,000-resample paired confidence intervals exceed zero; held-out target-response intervals include zero. Gains persist with category-disjoint and annotation-absent foils. Margin improvement differs from sign repair: COCO B/16 negative-margin prevalence falls only from 74.04% to 73.89%, while switched-image bbox precision falls by 4.95 percentage points on average. An independent four-setting MPS rerun provides direct paired margin gains over Mean and Max-.1, with small gains over the latter and locality trade-offs. Reranking demonstrates sparse local margin improvement within frozen candidates, not broad failure elimination or loss-free explanations.

Index Terms— CLIP, explainability, interventions, model auditing, worst-foil specificity

1 INTRODUCTION

Frozen vision-language models such as CLIP [1] are often interpreted by ranking image regions according to the change in image–text similarity after an intervention. CCI [2], for example, clusters CLIP patch tokens into regions and combines cluster importance into a heatmap. We audit only its candidate region with maximum target-score drop (CCI-top1), not the complete heatmap. A large drop says that a region matters for the target; it does not establish that the effect is *specific* to the target. Removing generic object evidence can reduce a competing semantic foil even more, while bounding-box overlap remains high. Likewise, averaging over many easy

foils can hide one high-impact confounder. In the frozen archive, on COCO B/16, 41.18% of eligible images have bbox precision $\geq .5$ and positive aggregate mean-class margin, yet negative worst-foil margin. Among the 17,725 screen-passers, 11,411 (64.38%) fail the worst-foil audit.

We audit the declared foil-response vector of the *selected* frozen region and report target response, worst-foil specificity, and spatial locality separately. The contribution is the post-selection audit and controlled evidence of local margin improvement, not the max/CVaR operator itself [3]. COCOA [4] defines a corpus-versus-foil attribution objective; CASE [5] diagnoses class-insensitive saliency and builds contrastive explanations. Here the object is a fixed selected region’s intervention-response vector: we test what locality and mean-class screens miss, then probe improvement with selection separated from held-out audit. Related CLIP attribution [6, 7, 8], class/concept contrasts [9, 10, 11], and caption foils [12] motivate this distinction. Image Description Enhanced CLIP-Adapter (IDEA) [13] combines visual features with image descriptions for few-shot classification. Our audit instead examines target specificity of selected regions with the model held fixed.

Reranking is a controlled probe of local improvement, not the principal algorithmic claim. Negative relative specificity does not imply that the region provides no faithful evidence for the target: evidence may support several classes.

2 WORST-FOIL AUDIT AND TARGET-TOLERANT RERANKING

For checkpoint m , $s_m(x, c) = \tau_m \langle z_m(x), e_m(c) \rangle$ uses unit-normalized embeddings and the checkpoint’s learned $\tau_m = \exp(\text{logit_scale}_m)$, before softmax. For candidate region R and concept prompt c , define

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c), \quad (1)$$

where $\mathcal{I}$ is a frozen internal intervention. Our single-region CCI baseline selects

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t), \quad (2)$$

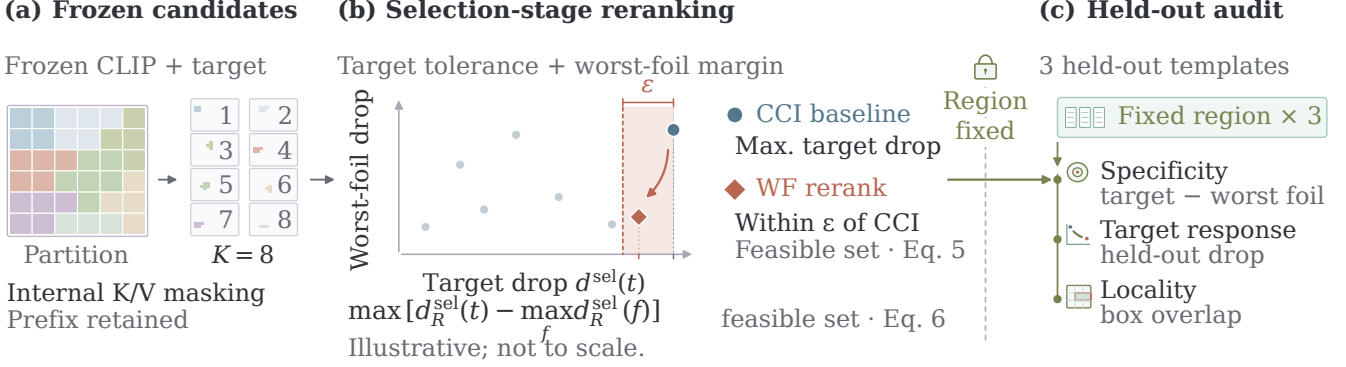


Figure 1: Selection and held-out audit. Frozen CCI candidates are selected or reranked only from selection-stage tensors. The region is fixed before held-out specificity, target-response, and locality evaluation. The candidate-response diagram is illustrative and not to scale.

using only the selection prompt. CCI-top1 denotes this baseline; the CCI subscripts and figure labels retain this shorthand. For non-target foils $\mathcal{F}_t$, a worst-foil margin compares the target response with the strongest foil response.

Matched held-out estimands. Selection uses “a photo of a {category}”. Held-out evaluation uses “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. Let $e_h(c)$ denote the stored unit-norm text embedding for held-out prompt h , and $n_c = \|H^{-1} \sum_h e_h(c)\|_2$. We apply the same class-specific normalization rule, $\tilde{d}_h(c) = d_h(c)/n_c$ and define

$$P_{\max}^{\text{pm}}(R, t) = H^{-1} \sum_h [\tilde{d}_h(t) - \max_f \tilde{d}_h(f)],$$

$$P_{\max}^{\text{agg}}(R, t) = H^{-1} \sum_h \tilde{d}_h(t) - \max_f H^{-1} \sum_h \tilde{d}_h(f). \quad (3)$$

Thus $P_{\max}^{\text{pm}} \leq P_{\max}^{\text{agg}}$: the former permits a different hardest foil per template. Both are ontology-relative specificity measures. For the mean-class screen, write $\bar{d}_c = H^{-1} \sum_h \tilde{d}_h(c)$. The archived screen uses $P_{\text{mean,all}} = \bar{d}_t - |\mathcal{C}|^{-1} \sum_{c \in \mathcal{C}} \bar{d}_c$ over all dataset classes, equally weighted, on these same held-out responses. Excluding the target gives $P_{\text{mean,foil}}$, with $P_{\text{mean,all}} = (|\mathcal{C}| - 1)P_{\text{mean,foil}}/|\mathcal{C}|$; their positive screens coincide. This class average differs from prompt-mean.

Selection-stage target tolerance. For each candidate define

$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_t} d_R^{\text{sel}}(f). \quad (4)$$

Given tolerance ϵ , form

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

and rerank only this feasible set:

$$R_{\text{WF}}(\epsilon) = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

Because $R_{\text{CCI}} \in \mathcal{R}_\epsilon$, the feasible set is nonempty; any switch loses at most ϵ of the *selection-stage* target drop. This is not

a held-out target guarantee. Equation 6 reads only selection-stage tensors, and held-out prompts are used only after the region is fixed. We use $\epsilon = .02$ as a fixed operating point, not an optimized value. Its cosine-scale tolerance is $.02/\tau_m$, not necessarily equal across checkpoints. Figure 1 summarizes the separation.

3 EXPERIMENTAL PROTOCOL

Models, intervention, and endpoints. We use $K = 8$ frozen CCI regions on COCO val2014 [14] (27,708 eligible images) and VOC2007 test [15] (1,943), with OpenAI CLIP B/16 and B/32. Reranking changes only which frozen candidate is selected. Each image has one target: the largest-area annotated instance’s category (COCO annotation area; VOC box area). Eligibility requires another annotated category and target-instance area between 2% and 80% of the image; COCO crowd annotations are excluded. Full foils comprise every other dataset category (79 for COCO, 19 for VOC). The intervention blocks selected patch key/value columns in every visual-transformer block and head by adding $-\infty$ before attention softmax, while retaining the prefix token. Cached patch features are clustered by K-means ($n_{\text{init}} = 3$, 50 iterations, seed $1701 + i$ for cache index i). BBox precision is the fraction of selected patch centers in the union of transformed boxes of the target category, not only its largest instance, and is reported separately because perturbation metrics depend on the intervention operator [16]. Ties select the first exact argmax.

Held-out controls and uncertainty. Selection uses one template and evaluation uses the three held-out templates in Eq. 3. Cross-foil controls use ten fixed seeds 1701, 2701, $\dots$, 10701 to permute categories into equal alternating folds. Each direction selects using one fold and audits on the other, excluding the target from both. The 20 seed–direction deltas are averaged within each image before paired unique-image resampling; annotation-absent evaluation removes categories annotated as present without claiming semantic absence. Fixed 5/19-foil subsets probe cardinality and composition jointly. Intervals use 10,000 paired unique-image bootstrap resamples; the

Table 1: Frozen archive: CCI-top1 vs. WF at $\epsilon = .02$. $P = P_{\max}^{\text{agg}}$; brackets are paired 95% CIs; “Neg.” is baseline→WF negative-margin rate.

Setting	P_{CCI}	ΔP	Δd_t	Δbbox	Sw.	Neg.
COCO B/16	-.342	+0.0150 [+.0131,+0.0169]	+0.00005 [-.00034,+0.00044]	-.00118 [-.00199,-.00036]	2.39%	74.04→73.89%
COCO B/32	-.304	+0.0108 [+.00945,+0.0122]	-.00010 [-.00046,+0.00027]	-.00080 [-.00162,+0.00002]	2.21%	73.91→73.75%
VOC B/16	+.168	+0.0118 [+.00677,+0.0183]	-.00085 [-.00227,+0.00059]	-.00340 [-.00652,-.00034]	2.57%	58.16→57.59%
VOC B/32	+.186	+0.00867 [+.00441,+0.0143]	-.00069 [-.00184,+0.00040]	+0.00023 [-.00288,+0.00341]	1.96%	55.89→55.58%

Table 2: Local rerun CCI-top1 screen audit. A : bbox $\geq .5$ and mean-foil margin > 0 ; F : worst-foil margin < 0 . Rates and conditional 95% CIs in %.

Setting	$\Pr(A)$	$\Pr(A \cap F)$	$\Pr(F A)$
COCO B/16	63.97	41.18	64.37 [63.67,65.05]
COCO B/32	64.41	41.72	64.78 [64.07,65.47]
VOC B/16	62.22	26.30	42.27 [39.45,45.05]
VOC B/32	63.15	25.99	41.16 [38.40,43.91]

archive fixes the base seed at 1701 and records deterministic setting/metric-specific seeds. Only four provenance-valid OpenAI settings enter the direct reranking analysis; two historical LAION settings are excluded because their text-normalization provenance is unresolved.

Independent local rerun. We regenerated candidates and responses on a MacBook M4 Pro using PyTorch 2.8.0 MPS, the same declared OpenAI models, splits, templates, $K = 8$ and $\epsilon = .02$. Tables 2, 3 and 6 report this rerun; Table 1, foil/tolerance controls and Figures 2–3 retain frozen-archive results. Rerun intervals are 10,000-draw percentile image bootstraps with recorded metric-specific seeds. Direct comparisons resample same-image differences; conditional failure jointly resamples numerator and denominator, and switch intervals resample switched images. Intervals are pointwise, without multiple-comparison adjustment.

4 RESULTS

Worst-foil debt. In the frozen archive, COCO/OpenAI B/16 CCI-top1 has normalized $P_{\max}^{\text{agg}} = -.342$ [−.357, −.327], and 74.04% [73.56, 74.61]% of selected regions have negative aggregate margin. Its held-out raw target response is .939 [.924, .953] and bbox precision .679 [.674, .683]. These are full-sample means, not estimates within the negative-margin subset. In that archive’s negative-margin subset, mean raw target drop is .481 and 90.40% have positive target drop. Table 2 adds the independent four-setting screen audit. Rerun COCO

B/16 has 11,410 failures among 17,726 screen-passers; the frozen counts remain 11,411/17,725.

Sparse margin improvement. Table 1 gives the direct baseline comparison. Normalized aggregate and prompt-mean deltas are positive with CIs above zero in all four settings; the corresponding prompt-mean deltas are +.0150, +.0108, +.0119, and +.00867. Held-out target-response intervals include zero in all four; this does not establish equivalence or a no-loss guarantee. Locality is not uniformly preserved: B/16 on COCO and VOC shows small bbox decreases, while B/32 intervals include zero.

The gain is not merely selector–endpoint reuse. Figure 2 shows that the gain persists under category-disjoint cross-foil and annotation-absent evaluation. Pooled normalized aggregate gains are +.01052 [+.00901, +.01198], +.00728 [+.00629, +.00846], +.00705 [+.00296, +.01140], and +.00438 [+.00131, +.00784] for COCO B/16, COCO B/32, VOC B/16, and VOC B/32. Restricting evaluation further to annotation-absent foils remains positive on the full eligible image population: +.00939, +.00697, +.00809, and +.00443, with all paired CIs above zero. These controls support gain persistence, not baseline failure-rate persistence under the same restrictions.

Direct same-feasible-set comparisons. Table 3 reports direct rerun WF-minus-alternative differences. Mean maximizes $d_t^{\text{sel}} - \text{mean}_f d_f^{\text{sel}}$ and Max-.1 maximizes $d_t^{\text{sel}} - .1 \max_f d_f^{\text{sel}}$ over the same $\mathcal{R}_\epsilon$. All eight pointwise margin CIs exceed zero, but gains over Max-.1 are small. Relative to Mean, bbox declines on COCO B/16, COCO B/32 and VOC B/16 with CIs below zero. Relative to Max-.1, bbox CIs include zero in all settings, while COCO B/16 raw target response decreases by .000099 [−.000192, −.000010]. These results do not establish loss-free superiority.

Why switching is sparse. In the local rerun, singleton feasible sets account for 95.29/95.60/95.21/95.27% of COCO B/16, COCO B/32, VOC B/16 and VOC B/32 images. The counts with 2/3/4 candidates are 1215/81/8, 1174/43/2, 88/5/0 and 89/3/0; none have more than four. COCO B/16 permits an alternative in 1,304/27,708 images, and 663/1,304 (50.84%) switch. The frozen counterpart is 1,303.

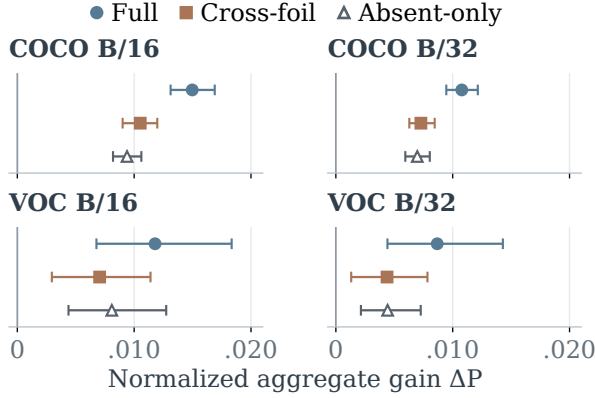


Figure 2: Held-out normalized aggregate gains at $\epsilon = .02$ for full, category-disjoint cross-foil, and annotation-absent evaluation; bars are paired 95% CIs.

Table 3: Local rerun direct paired $\Delta P_{\max}^{\text{agg}}$, multiplied by 10^3 ; pointwise 95% CIs. Each column subtracts the named alternative from WF.

Setting	WF-Mean	WF-Max-.1
COCO B/16	12.461 [10.838,14.193]	0.245 [0.131,0.372]
COCO B/32	9.343 [7.970,10.807]	0.241 [0.088,0.418]
VOC B/16	4.480 [2.332,7.019]	0.507 [0.025,1.114]
VOC B/32	3.184 [1.450,5.126]	0.418 [0.044,0.907]

Table 4: Frozen archive: COCO foil-count sensitivity. “Neg.” is baseline negative-margin rate; ΔP is normalized aggregate gain.

Setting	Neg.@5	Neg.@19	Neg.@all	ΔP_5	ΔP_{19}
COCO B/16	45.8%	63.5%	74.0%	+.00872	+.01074
COCO B/32	45.0%	62.8%	73.9%	+.00841	+.00947

The effect also reduces the breadth of foil violations. As a secondary held-out diagnostic, let $\bar{d}(c) = H^{-1} \sum_h \bar{d}_h(c)$ for the already fixed region. We report V_{rate} , the fraction of foils with $\bar{d}(f) > \bar{d}(t)$, and $V_{\text{mass}} = |\mathcal{F}_t|^{-1} \sum_{f \in \mathcal{F}_t} [\bar{d}(f) - \bar{d}(t)]_+$, averaging over all foils, including zeros. On COCO/OpenAI B/16, reranking at $\epsilon = .02$ changes V_{rate} from .18973 to .18751 and V_{mass} from .05878 to .05708. Violation breadth therefore also decreases descriptively.

Foil-cardinality sensitivity. Adding foils can increase the maximum response, but the fixed subsets in Table 4 vary cardinality and composition jointly. The reranking gain stays positive at both matched counts, so the effect is not created solely by evaluating against a larger foil set.

Cross-template consistency. Across the three held-out phrases, mean pairwise hardest-foil identity agreement is 59.80%. All three agree on the exact category in 46.36% of images and on its stored semantic family in 53.69%; the latter two use the

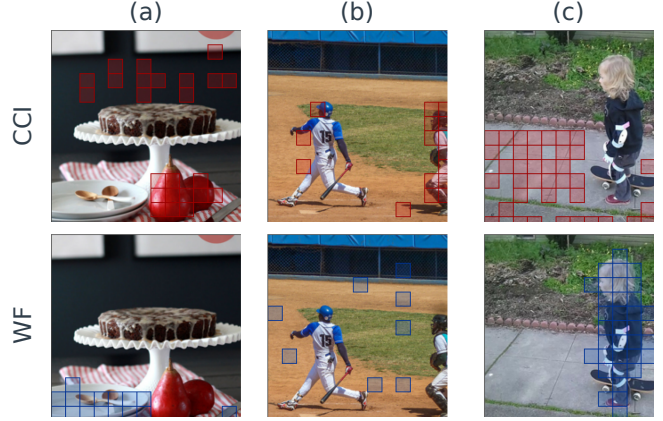


Figure 3: Archived rule-selected COCO/OpenAI B/16 cases at $\epsilon = .02$ (top/bottom: CCI/WF); ΔP is normalized aggregate gain: (a) locality gain, dining table, $\Delta P = +4.58$; (b) specificity–locality trade-off, person, $\Delta P = +0.78$; (c) counterexample, person, $\Delta P = -0.98$. Bbox precision: .46 $\rightarrow$ 1.00, .81 $\rightarrow$.11, and .02 $\rightarrow$.88, respectively.

Table 5: Frozen archive: COCO/B16 tolerance sensitivity; ΔP is full-sample gain, while $\Delta P \mid \text{sw}$ and $\text{Pr}(+)$ condition on switches.

ϵ	Sw.	ΔP	$\Delta P \mid \text{sw}$	$\text{Pr}(+)$
.01	1.23%	+.00791	+.642	90.9%
.02	2.39%	+.01497	+.626	92.2%
.05	6.24%	+.03906	+.626	91.6%
.10	11.68%	+.07643	+.654	92.0%
.20	21.87%	+.14479	+.662	92.9%

same all-three criterion, not the pairwise criterion. Families follow the frozen metadata mapping, e.g., vehicle and animal.

Tolerance sensitivity. Table 5 uses the same frozen candidates and selection tensors; $\epsilon = .02$ was fixed rather than selected from it. As tolerance grows, switches increase while $\Delta P \mid \text{sw}$ stays stable. Full-sample target/bbox deltas are (+.00005, −.00118) at .02, (−.00049, −.00223) at .05, and (−.01396, −.01627) at .20, exposing the target–specificity trade-off.

Switch-conditioned outcomes. Table 6 uses rerun switched-image vectors directly. VOC B/16 loses 13.20 bbox percentage points [−24.62, −1.80]. COCO B/16 loses 4.95 points [−8.33, −1.45], with median zero and quartiles −.3242, +.1137. Raw target-response intervals include zero in all four switch subsets; this is not an equivalence result. Margin improvement occurs in 611/663, 532/614, 46/50 and 35/38 switches, whereas sign repair occurs in only 42/663, 48/614, 11/50 and 6/38.

Cases and uncertainty. Figure 3 retains archived rule-selected cases and does not estimate frequencies. Rerun outputs pass strategy/candidate-array consistency checks for all 59,302 image–model records. Small CPU/MPS checks

Table 6: Local rerun switch-conditioned WF–CCI outcomes at $\epsilon = .02$. Raw target and bbox means with 95% switched-image CIs; bbox multiplied by 100 (percentage points).

Setting	n_{sw}	Δd_t	$100\Delta \text{bbox}$
COCO B/16	663	0.002 [-0.013,0.018]	-4.95 [-8.33,-1.45]
COCO B/32	614	-0.004 [-0.021,0.012]	-3.61 [-7.45,0.05]
VOC B/16	50	-0.032 [-0.086,0.020]	-13.20 [-24.62,-1.80]
VOC B/32	38	-0.035 [-0.091,0.021]	1.16 [-15.00,17.26]

cover only 1–2 images per setting; they do not establish full-run backend equivalence.

Interpretation and scope. Rerun and archive are separate evidence sets. COCO B/16 image IDs align, but three images change screen/failure status and ten change CCI region labels. Some response changes exceed threshold rounding; frozen candidate tensors are unavailable for causal attribution. COCO B/32 switches 614 times versus 613 in the archive, whose missing per-image fields prevent localization. These differences are not attributed to MPS or treated as exact reproduction. Inference is conditional on eligible images, templates, foils, candidates and intervention. Restricted-foil conditional failure rates and robustness to new candidate or intervention configurations remain untested.

Conclusion. CCI-top1 auditing exposes specificity failures and sparse margin improvement. Independent reruns support positive paired margin gains over Mean and Max-.1 under the tested protocol, with small gains over Max-.1 and measurable locality costs. Target response, locality and sign repair remain distinct endpoints.

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